

Conditional Distribution Matching with Discrete Inputs: Prompt Optimization via Twisted SMC

Working notes — experimental summary

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July 2026

Abstract

We extend Conditional Distribution Matching (CDM) to *discrete* inputs: find a prompt x^* such that the induced conditional $\mathcal{P}(Y \mid X=x^*)$ of a frozen text-to-image model matches a target distribution $\mathcal{G}(Y)$. Gradients with respect to x do not exist, so we replace MLGD’s gradient correction with *selection*: a twisted Sequential Monte Carlo sampler over a masked (absorbing) discrete diffusion prior, in which the distributional loss enters only as a scalar. Three additions make the method work at scale: (i) *cross-modal tilted proposals* — a CLIP-text surrogate nominates and ranks candidate tokens so that rare target-relevant words are actually proposed; (ii) a *remasking corrector* that lets the chain revise committed tokens; and (iii) a non-distilled conditional sampler, which we show is *necessary* for multimodal targets because adversarially distilled models (SDXL-Turbo) collapse linguistic ambiguity to a single mode. With the full stack, a multimodal target defined by the prompt “an old man or young woman smiling” is recovered from a 75%-masked corruption in 5/6 runs up to the estimation noise floor of the objective, with induced demographics matching the target.

1 Problem

CDM (*main paper*) seeks an input whose induced conditional matches a user-specified target distribution. Here the input space is discrete: x is a token sequence (a prompt), the conditional model is a frozen text-to-image generator f_ϕ , and we seek

$$x^* = \arg \min_x \mathcal{L}(x), \quad \mathcal{L}(x) = \text{MMD}^2(\{\text{CLIP}(y_j)\}_{j=1}^{n_{\text{cond}}}, S_{\mathcal{G}}), \quad y_j \sim f_\phi(x, \cdot), \quad (1)$$

where $S_{\mathcal{G}}$ is a fixed set of CLIP embeddings of target images ($n_{\text{cond}}=8$ with common random numbers so $\mathcal{L}$ is a deterministic, memoizable function of x). As in the main paper we target the tempered posterior $\mathcal{Q}_\beta(x) \propto \mathcal{P}(x) e^{-\beta \mathcal{L}(x)}$.

2 Method

2.1 Setup and notation

Let V be a vocabulary of A word tokens and $\mathbf{m} \notin V$ a distinguished MASK symbol. A *canvas* is a fixed-length sequence $x = (x^1, \dots, x^L) \in (V \cup \{\mathbf{m}\})^L$; we call a canvas *decoded* if no position carries $\mathbf{m}$. The prompt handed to the image model is $\text{dec}(x)$: a fixed natural-language prefix (e.g.

*Notes compiled from the `discrete-x-smc` experimental branch; all runs on a single L40S.

“a photo of”) followed by the canvas words. V is the intersection of the masked LM’s alphabetic word tokens with CLIP’s single-token words ($A \approx 18\text{k}$ for BERT), so that one canvas slot is exactly one CLIP text token — this alignment is what makes the cross-modal proposals of §2.5 possible.

The loss oracle. For a decoded canvas, $\mathcal{L}$ in (1) is computed by generating $n_{\text{cond}}=8$ images $y_j \sim f_\phi(\text{dec}(x), \cdot)$ with *common random numbers* (the same n_{cond} initial latents for every prompt), embedding them with CLIP ViT-L/14, and evaluating the biased (V-statistic) squared maximum mean discrepancy against the target set $S_G = \{s_1, \dots, s_{n_{\text{targ}}}\}$, $n_{\text{targ}}=64$:

$$\text{MMD}^2(E, S_G) = \frac{1}{n^2} \sum_{j,j'} k(e_j, e_{j'}) + \frac{1}{m^2} \sum_{l,l'} k(s_l, s_{l'}) - \frac{2}{nm} \sum_{j,l} k(e_j, s_l), \quad k(u, v) = e^{-\gamma \|u-v\|^2}, \quad (2)$$

with the RBF bandwidth set by the median heuristic on the target set, $\gamma^{-1} = \text{median}\{\|s_l - s_{l'}\|^2\}_{l < l'}$. Common random numbers make $\mathcal{L}$ a *deterministic function of the prompt string*, so evaluations are memoized and repeated queries are free. Nothing downstream uses gradients of $\mathcal{L}$; it is consumed strictly as a scalar (a “loss oracle”).

Goal. Sample from the tempered posterior

$$\mathcal{Q}_\beta(x_0) \propto \mathcal{P}(x_0) e^{-\beta \mathcal{L}(x_0)}, \quad (3)$$

where $\mathcal{P}$ is a prior over decoded canvases (defined next) and $\beta > 0$ is an inverse temperature ($\beta=200$ throughout). As $\beta \rightarrow \infty$, $\mathcal{Q}_\beta$ concentrates on the minimizers of $\mathcal{L}$ within the support of $\mathcal{P}$; the prior’s role is to restrict the search to fluent English rather than adversarial token strings.

2.2 The prior: masked (absorbing) discrete diffusion

Forward process. Discrete times $t = 0, 1, \dots, T$ ($T=16$). Given a decoded canvas x_0 , the forward process masks each position independently:

$$q(x_t | x_0) = \prod_{i=1}^L q(x_t^i | x_0^i), \quad q(x_t^i = \mathbf{m} | x_0^i) = \frac{t}{T}, \quad q(x_t^i = x_0^i | x_0^i) = 1 - \frac{t}{T}, \quad (4)$$

so x_T is all-masked and information is destroyed by *absorption* only — a token is either intact or masked, never substituted.

Denoiser. The learned object is the conditional $p_\theta(x_0^i = v | x_t)$ for masked positions i , i.e. “which word sits under this mask given the visible context”. This is *exactly* the masked-language-modelling objective, so an off-the-shelf MLM (**bert-base-uncased**) is a legitimate denoiser without any training; we also use LLaDA-8B, a diffusion LM trained explicitly with this objective. Both expose the same interface: per-position marginals $p_\theta(\cdot | x_t) \in \Delta(V)$ over the restricted vocabulary.

Reverse kernel. One reverse step $r_\theta(x_{t-1} | x_t)$ acts independently per position: an *unmasked* position is copied unchanged; a *masked* position unmask with probability $1/t$, and if it unmask, its word is drawn from the denoiser marginal truncated to its top- k ($k=50$, renormalized — standard top- k decoding; this truncation modifies the prior we sample, not the correctness of the weighting below). The rate $1/t$ makes the reverse marginals consistent with the forward ones: a position masked at time t survives to time $t-j$ with probability $\prod_{s=t-j+1}^t (1 - \frac{1}{s}) = \frac{t-j}{t}$, and at $t=1$ everything unmask, so the chain always terminates decoded. The prior $\mathcal{P}(x_0)$ in (3) is *defined* as the law of x_0 under running r_θ from the all-masked canvas.

2.3 Guidance: a Feynman–Kac model solved by twisted SMC

We cannot evaluate $\mathcal{P}(x_0)$ or its gradients; what we *can* do is simulate the reverse chain and query the scalar oracle $\mathcal{L}$. This is precisely the setting of Feynman–Kac (FK) models: reweight a simulable Markov chain by potential functions so that its *final* state is distributed according to a twisted target.

Optimal twist. Define, for each time t , the value function (“twist”)

$$h_t(x_t) = \mathbb{E}\left[e^{-\beta\mathcal{L}(x_0)} \mid x_t\right], \quad (5)$$

the expectation being over the remainder of the reverse chain $x_{t-1}, \dots, x_0 \sim r_\theta$ started at x_t . Two properties matter:

- (i) *Terminal exactness*: a decoded canvas has no remaining randomness, so $h_0(x_0) = e^{-\beta\mathcal{L}(x_0)}$ exactly.
- (ii) *Tower property*: by the law of total expectation, $\mathbb{E}_{r_\theta}[h_{t-1}(x_{t-1}) \mid x_t] = h_t(x_t)$.

The FK path targets. For the chain started at t_0 (see §2.4 for the start distribution μ), define unnormalized path densities

$$\gamma_t(x_{t_0:t}) = \mu(x_{t_0}) \prod_{s=t}^{t_0-1} r_\theta(x_s \mid x_{s+1}) \cdot h_t(x_t), \quad t = t_0, \dots, 0. \quad (6)$$

At $t=0$, by (i), the x_0 -marginal of γ_0 is proportional to $\mathcal{P}(x_0) e^{-\beta\mathcal{L}(x_0)} = \mathcal{Q}_\beta(x_0)$ — the target (3). Sequential importance sampling on this sequence, with the prior kernel r_θ as proposal, gives the incremental weight

$$\alpha_t(x_t, x_{t-1}) = \frac{\gamma_{t-1}(x_{t_0:t-1})}{\gamma_t(x_{t_0:t}) r_\theta(x_{t-1} \mid x_t)} = \frac{h_{t-1}(x_{t-1})}{h_t(x_t)}, \quad (7)$$

because everything except the twists cancels. By (ii) these weights have conditional mean one, and with the *exact* twist they are the minimum-variance choice (the “optimal twist” of the SMC literature).

The algorithm. $N=128$ particles $\{x^n\}$ with log-weights $\{\log w^n\}$:

1. **Init** ($t = t_0$): draw $x^n \sim \mu$, set $\log w^n = \log \hat{h}_{t_0}(x^n)$.
2. **Propagate**: $x_{t-1}^n \sim r_\theta(\cdot \mid x_t^n)$ (or the tilted proposal of §2.5, followed by the corrector of §2.6).
3. **Reweight**: $\log w^n += \log \hat{h}_{t-1}(x_{t-1}^n) - \log \hat{h}_t(x_t^n)$ [$+\log r_\theta - \log q$ if tilted].
4. **Resample**: with normalized weights $W^n = e^{\log w^n} / \sum_m e^{\log w^m}$, compute the effective sample size $\text{ESS} = 1 / \sum_n (W^n)^2$ (the number of “useful” particles: N when weights are uniform, 1 when one particle carries all mass). If $\text{ESS} < N/2$, resample systematically (duplicate high-weight particles, kill low-weight ones, preserving expectations) and reset $\log w^n = 0$.
5. Loop $t \rightarrow t - 1$ until $t=0$; report the best decoded particle by $\mathcal{L}$ (optimization view) or the weighted ensemble (sampling view).

Plug-in twist estimators. The exact h_t of (5) is intractable; we substitute a plug-in $\hat{h}_t$ built from the denoiser’s factorized posterior $\tilde{p}_\theta(x_0 | x_t) = \prod_{i: x_t^i = \mathbf{m}} p_\theta(x_0^i | x_t)$ (independent fills — itself an approximation, since true fills are correlated):

$$\textbf{mode: } \hat{x}_0(x_t)^i = \begin{cases} x_t^i & x_t^i \neq \mathbf{m} \\ \arg \max_v p_\theta(v | x_t) & \text{else} \end{cases}, \quad \hat{h}_t(x_t) = e^{-\beta_t \mathcal{L}(\text{dec}(\hat{x}_0))}, \quad (8)$$

$$\textbf{sampled: } x_0^{(j)} \sim \tilde{p}_\theta(\cdot | x_t), \quad j = 1:n_{\text{dec}}, \quad \hat{h}_t(x_t) = \frac{1}{n_{\text{dec}}} \sum_{j=1}^{n_{\text{dec}}} e^{-\beta_t \mathcal{L}(\text{dec}(x_0^{(j)}))}. \quad (9)$$

Mode costs one oracle call per particle per step; sampled costs n_{dec} (we use 4–8) and is unbiased for the twist *under the factorized posterior* — the residual bias is exactly the factorization. Crucially, plug-in error cannot corrupt the final target: the reweighting telescopes,

$$\sum_{t=1}^{t_0} [\log \hat{h}_{t-1} - \log \hat{h}_t] + \log \hat{h}_{t_0} = \log \hat{h}_0 = -\beta \mathcal{L}(x_0), \quad (10)$$

and $\hat{h}_0$ is exact by terminal exactness (i). Intermediate twist error therefore only shifts *where resampling focuses compute* (variance / efficiency), not the distribution the weights ultimately encode — the standard consistency argument for twisted SMC with approximate twists.

β -annealing. We use the schedule $\beta_t = \beta(T - t)/T$: zero guidance at $t=T$, full β at $t=0$. Rationale: early in the chain most positions are masked, so $\hat{x}_0$ is dominated by the denoiser’s unconditional guesses and $\mathcal{L}(\hat{x}_0)$ is nearly uninformative about the particle — full-strength weighting there just resamples on noise and collapses diversity before any signal exists. Since $\beta_0 = \beta$, the final target is still $\mathcal{Q}_\beta$. Annealing was essential in every setting we tested (without it, the toy benchmark drops from 5/5 to 0/5 recovery).

2.4 SDEdit-style initialization (editing and recovery)

Rather than starting all-masked ($\mu = \delta_{\text{all-m}}, t_0 = T$), every particle starts from a *source* canvas x_{src} with each position independently re-masked w.p. t_0/T — an exact draw from the forward corruption $q(x_{t_0} | x_{\text{src}})$ — and the reverse chain runs from t_0 ($t_0/T = 0.75$ throughout). Because particles now differ at initialization, the correct initial FK weight is the twist itself, $\log w^n = \log \hat{h}_{t_0}(x^n)$ (for the identical all-masked start this is a constant and reduces to uniform weights, so from-scratch behavior is unchanged). The *recovery* task sets $x_{\text{src}} = x_{\text{true}}$ (denoise your way back); the *editing* task sets it to a semantically different prompt.

2.5 Cross-modal tilted proposals

Bootstrap SMC proposes from the prior and lets weights select — but selection cannot select what is never proposed. At vocabulary scale the denoiser’s marginals concentrate on generic words; target-relevant rare words (“rain”) sit at rank $\sim 5,000$ of 18k and are effectively never drawn by top- k sampling. We therefore *tilt* the proposal toward the target using a differentiable surrogate, and pay for it exactly in the weights.

Surrogate. Let $\bar{s}_G = \frac{1}{|\bar{S}_G|} \sum_l s_l / \|\cdot\|$ be the normalized mean target embedding and

$$\mathcal{L}_{\text{sur}}(x) = 1 - \cos(\text{CLIP}_{\text{text}}(\text{dec}(x)), \bar{s}_G), \quad (11)$$

a *text-side* proxy for the image-side loss: no image generation, one text-encoder forward. CLIP’s shared text–image embedding space is what makes comparing a *prompt* embedding to *image* embeddings meaningful; this is the cross-modal step.

Step 1 — gradient nomination. Write the canvas one-hot: slot i carries $q_i \in \Delta(V)$, and the CLIP text embedding of slot i is $q_i^\top E$ with $E \in \mathbb{R}^{A \times d}$ the token embedding matrix. Relaxing q_i to the simplex makes $\mathcal{L}_{\text{sur}}$ differentiable in $(q_1, \dots, q_L)$; *one* backprop at the current mode-decode $\hat{x}_0$ yields $g_{i,v} = \partial \mathcal{L}_{\text{sur}} / \partial q_{i,v}$ for *all* slots i and words v simultaneously — a first-order estimate of the effect of substituting word v into slot i (Gibbs-with-Gradients style). This gradient is used *only to nominate*: candidate pool $C_i = \text{top-}k(p_\theta(\cdot | x_t)) \cup \text{top-}k(-g_{i,\cdot})$, $k = 50$.

Step 2 — exact scoring. For each nominee $v \in C_i$, evaluate $\mathcal{L}_{\text{sur}}(\hat{x}_0^{i \leftarrow v})$ — where $\hat{x}_0^{i \leftarrow v}$ denotes the mode decode with slot i replaced by word v — *exactly*, with one batched CLIP-text forward over the pool ($|C_i| \leq 100$ prompts). No Taylor approximation survives into the numbers used for sampling.

Step 3 — tilted fill and exact weight correction. A position that unmask draws its word from

$$q_i(v) \propto p_\theta(v | x_t) e^{-\lambda \beta_{t-1} \mathcal{L}_{\text{sur}}(\hat{x}_0^{i \leftarrow v})}, \quad v \in C_i, \quad (12)$$

with tilt strength $\lambda=1$. Changing the proposal from r_θ to q changes the denominator of (7), so each tilted fill contributes the exact correction $\log p_\theta(v | x_t) - \log q_i(v)$ to the particle’s log-weight (these accumulate over filled positions within the step). Consequence: *a biased surrogate can bias only efficiency, never the target* — if $\mathcal{L}_{\text{sur}}$ nominates junk, the weights pay for it and resampling discards it. The one caveat is support: q puts no mass outside C_i , comparable in kind to the top- k truncation already present in the baseline kernel.

A pure first-order tilt over the full vocabulary ($q(v) \propto p_\theta(v) e^{-\lambda \beta g_{i,v}}$, the published D-CBG recipe) *fails* here — loss increases with tilt strength — because the Taylor error at the distance of a full one-hot swap is larger than the signal. Nominate-with-gradients, score-exactly is what works. Since the surrogate lives entirely in text space, its overhead at image scale is $\sim 25\%$ of wall-clock.

2.6 Remasking corrector

Absorbing diffusion can never revise an unmasked token — whatever is committed (from the source prompt or an early bad fill) survives verbatim unless re-masked. We observed exactly this failure: corrupted source tokens surviving into final answers. The corrector inserts, after each propagation step, an independent re-masking of every *unmasked* position with probability

$$\sigma_t = \sigma \frac{t-1}{T}, \quad \sigma = 0.2, \quad (13)$$

which decays to $\sigma_1 = 0$ so the chain still terminates decoded. The composite kernel (unmask-fill $\circ$ re-mask) is treated as *the* proposal kernel of the FK model: the telescoping identity of §2.3 is agnostic to which kernel generated the path, so the final weights still encode $e^{-\beta \mathcal{L}(x_0)}$ against the law of x_0 under the *corrector-augmented* chain. That law is a (slightly) different effective prior

than pure absorbing diffusion — ReMDM-style schedules choose σ_t to preserve the prior marginals exactly; our fixed decaying schedule treats this as an approximation, which we accept in exchange for reversibility. Empirically the corrector took the text-space rain benchmark from a 0.24 plateau to *exact recovery* ($\mathcal{L} = 0.0000$) and produced the compositional straddle of §3.4.

3 Experiments

All image-scale runs: SDXL (Turbo 1-step or base 12-step CFG 3) as f_ϕ , CLIP ViT-L/14 embeddings, $N=128$, $T=16$, $\beta=200$ annealed, seeds $\{0, 1, 2\}$, single L40S, minutes per run (mode estimator).

3.1 Validation ladder

Toy world (5-slot grammar, 3,750 prompts, exact denoiser, brute-forceable optimum): with β -annealing the *sampled* estimator recovers the generating prompt 5/5 seeds; *mode* 1/5 (its deterministic twist is biased and all failures collapse to the same wrong prompt). **At image scale the ranking flips:** mode $\geq$ sampled on every seed (e.g. rain pair: mean 0.454 vs 0.479) at 1/12 the cost — greedy decodes are coherent sentences, factorized samples are not. Mode is the working configuration.

3.2 Prompt editing (unimodal target)

Target $\mathcal{G}$: images of “a photo of a person standing in the rain” ($\mathcal{L}(x_{\text{true}}) = 0.072$); source “a photo of a man walking down the street” ($\mathcal{L} = 0.80$). Selection alone inserts rain semantics once in three seeds (best 0.393); tilted proposals make rain vocabulary appear in 2/3 seeds (best 0.373, “man in precipitation water”); both plateau because 5 free tokens cannot express composition — the target is *distant full-body* silhouettes. Widening the canvas to 10 tokens breaks the plateau: $\mathcal{L} = 0.189$ (10-token optimum 0.095), with generations visually interchangeable with the target. Across 18 baseline runs and 3 source–target pairs, untilted SMC reliably performs *high-prior* semantic swaps (man→woman: 6/6) and never inserts *low-prior* content words — the coverage bottleneck that the tilt fixes.

3.3 Distillation collapses ambiguity

Defining $\mathcal{G}$ as a 50/50 mixture of “old man smiling” / “young woman smiling” images: under SDXL-**Turbo**, all seeds converge to the identical single-mode prompt (“a photo of a woman smiling in the image”, induced $p(\text{male})=0.01$ vs target 0.49), and a probe over hand-crafted straddle prompts shows why: *no* prompt beats the single-mode floor; disjunctions *blend* attributes (old, mixed-gender people) rather than sampling modes; “a person smiling” induces $p(\text{male}) = 0.007$. Under non-distilled SDXL-**base** (12 steps, CFG 3) the picture inverts: disjunction prompts *beat* both components (0.086 vs 0.090/0.097) and “a person smiling” jumps to $p(\text{male}) = 0.56$. Re-running the mixture optimization with the base sampler, recovered prompts induce genuinely mixed demographics ($p(\text{male}) \in \{0.26, 0.78\}$ vs target 0.54). Because the SMC never differentiates through f_ϕ , this sampler swap is architecturally free — in contrast to MLGD-F, whose memory tractability requires the distilled model. *Selection-based guidance decouples CDM from distilled samplers.*

3.4 Headline: recovery of a multimodal target

Ground truth $x_{\text{true}} = \text{“a photo of an old man or young woman smiling”}$ (a 7-token disjunction the base sampler renders bimodally); $\mathcal{G} =$ its own 64 images; $\mathcal{L}(x_{\text{true}}) = 0.099$ is the finite-sample noise floor of the objective. Particles start from x_{true} with 75% of tokens masked (classic denoising setup).

prior	recovered prompt (best particle)	$\mathcal{L}$	$p(\text{male})$	$p(\text{old})$
<i>target (measured on $S_{\mathcal{G}}$)</i>		0.099	0.60	0.84
BERT	“some old woman or young man smiling”	0.083	0.44	0.85
BERT	“an old man or boy was smiling”	0.093	0.77	0.88
BERT	“an old boy or old woman smiling”	0.087	0.51	0.91
LLaDA-8B	“an old person or young person smiling” (2 seeds)	0.089	0.48	0.83
LLaDA-8B	“an elderly man smiling again showing the”	0.089	0.81	0.91

Table 1: Noised-truth recovery of the multimodal target. 5/6 runs recover the disjunction structure; all reach or beat the ground-truth prompt’s own empirical loss, i.e. they are statistically indistinguishable from the optimum (demographics computed on 64 *fresh* generations, so this is not overfitting to the target set). The trained diffusion LM (LLaDA) contributes fluency and cross-seed stability.

Editing from a semantically *far* source (“one person sitting quietly inside the office”) does not recover the disjunction (best 0.106 with corrector vs floor 0.099), but the corrector produced a qualitatively distinct solution: “a photo of her laughing with her father” — matching the mixture *compositionally* (both modes in one image, induced 0.55/0.60) rather than distributionally. CLIP-space MMD partially conflates the two; we flag this as a metric-level ambiguity.

3.5 Equal-compute baselines

Each baseline receives ~ 250 unique loss-oracle evaluations per seed, matched to (or above) the guided runs’ consumption, and shares the memoized loss. *SDEdit best-of-B*: $B=256$ independent *unguided* denoisings from the same noised source (same prior, same t_0 , same top- k reverse kernel — everything except guidance), keep the lowest-loss decode. *Random search*: $B=256$ uniform-random canvas fills.

task	method (equal oracle budget)	$\mathcal{L}$ best (3 seeds)	disjunction found
recovery	random prompt search ($B=256$)	0.153–0.170	0/3
recovery	SDEdit best-of-256, BERT	0.091–0.098	3/3
recovery	SDEdit best-of-256, LLaDA	0.089–0.102	2/3
recovery	guided SMC (ours), BERT	0.083 –0.093	3/3
recovery	guided SMC (ours), LLaDA	0.089 (all seeds)	2/3
editing (far source)	SDEdit best-of-256, BERT	0.157–0.163	0/3
editing (far source)	guided SMC (ours)	0.106	compositional

Table 2: Equal-compute baselines. Unguided restarts are competitive when the corruption preserves part of the answer (noised-truth recovery: best-of-256 inherits $\sim 25\%$ of true tokens per candidate, and one of 256 gets lucky), and no better than random search when the source is semantically far — guidance accounts for the entire gap in the editing regime.

The two regimes give the honest claim: guidance pays *modestly* for recovery (per-seed best: guided mean 0.088 vs unguided 0.094; the population-level gap is large — unguided median 0.18 — which is what “guidance concentrates compute” looks like), and *decisively* for editing, where 256 unguided restarts from a far source are statistically indistinguishable from pure random prompt search (0.157–0.163 vs 0.153–0.170) while guided SMC reaches 0.106, $\sim 35\%$ better.

3.6 Ablations

One component removed per arm, everything else identical to the full stack (3 seeds each), on both tasks.

arm	recovery (floor 0.099)		editing, far source	
	$\mathcal{L}$ per seed	mean	$\mathcal{L}$ per seed	mean
full stack	.083/.093/.087	0.088	.128/.124/.106	0.119
– tilted proposals	.089/.087/.088	0.088	.141/.102/.114	0.119
– corrector	.084/.094/.087	0.088	.136/.115/.132	0.128
– β -annealing	.087/.087/.084	0.086	.135/.130/.132	0.132
unguided (best-of-256)	.091/.094/.098	0.094	.157/.163/.159	0.160

Table 3: Component ablations. Recovery needs only the twisted selection itself: every knockout matches the full stack, and the entire guided-vs-unguided gap is the twist. On editing, β -annealing and the corrector each contribute (without annealing all seeds collapse to single-mode prompts), while the tilt is neutral *on this target* — its vocabulary (old/man/woman/smiling) is already high-prior. The tilt’s demonstrated value is inserting *low-prior* content words (§“prompt editing”: rain vocabulary appears only with tilt).

A separate distilled-sampler arm on recovery is the sharpest collapse evidence we have: under SDXL-Turbo, 2/3 seeds recover x_{true} *verbatim* — yet even the true disjunction prompt induces $p(\text{male}) = 0.13$, and Turbo’s own rendering of the target set has $p(\text{male}) = 0.20$ vs 0.60 under the base sampler. The text is recoverable; the multimodal distribution is unreachable (cf. §3.3).

3.7 A rate-based alternative: training-free CTMC guidance

Twisted SMC is not the only way to consume a scalar oracle. Rate-ratio guidance (Nisonoff et al.; TFG-Flow) works on the continuous-time Markov chain directly: a *single* chain, no particles, no weights, with the predictor $f(x_1) = e^{-\beta\mathcal{L}(x_1)}$ reweighting each unmasking jump. For a masked coordinate d unmasking at time s , the guided destination law is

$$q_d(v) \propto \mathbb{E}_{x_1 \sim \tilde{p}_\theta(\cdot|x_s)}[f(x_1) \mathbf{1}\{x_1^d = v\}], \quad (14)$$

estimated either by **Monte Carlo** (K posterior decodes per step, f -weighted empirical law — assumption-free, and attractive precisely because it needs *only* the scalar oracle: no CLIP-text surrogate, no gradients, no shared embedding space), or in **closed form** under the relaxing assumption that f is log-linear across coordinates ($\log f = \sum_d \theta_d(x^d)$; then $q_d = \text{softmax}(\log p_\theta + \gamma \theta_d)$, one denoiser pass, zero sampling).

In the toy world the method matches SMC: MC rate guidance with $K=128$, $\gamma=10$, 16 restarts recovers the generating prompt 5/5 seeds at $\sim 9\times$ fewer raw oracle calls than the SMC toy configuration. The closed-form variant plateaus ($\mathcal{L} \approx 0.08$ vs floor 0.009) and *degrades* as γ grows: a distribution-matching loss couples coordinates by construction, so its additive projection is a poor surrogate — the assumption’s own scope caveat, measured.

At image scale the ranking inverts (same tasks, matched ~ 250 unique oracle calls per seed):

method	recovery (mean)	editing (mean)
twisted SMC (full stack)	0.088	0.119
unguided SDEdit best-of-256	0.094	0.160
CTMC rate ratio, MC $K=16 \times 4$ chains, $\gamma=10$	0.111	0.165
CTMC rate ratio, MC $K=16 \times 4$, $\gamma=3$	0.107	—
CTMC rate ratio, MC $K=64 \times 2$, $\gamma=10$	0.126	—
CTMC rate ratio, closed-form additive	0.134	—
CTMC rate ratio, MC + DFM stochasticity $\eta=3$	0.140	—
CTMC rate ratio, MC + DFM stochasticity $\eta=8$	0.136	0.143
CTMC rate ratio, MC + regret-guided remask, $\eta=8$	0.131	0.143

Table 4: Rate-ratio guidance vs twisted SMC at image scale (floor 0.099). Single-chain rate guidance trails even unguided restarts on recovery and adds nothing over unguided on editing; detailed-balance stochasticity (η) trades the two tasks against each other rather than closing the gap, and allocating the same churn by commit-time regret does not repair the trade-off.

Three observations pin down *why*. First, $\gamma=3$ vs $\gamma=10$ barely differ — guidance strength is not the binding constraint. Second, reallocating the budget toward sharper per-step estimates ($K=64$, 2 chains) makes things *worse* than $K=16$ with 4 chains: better local twist estimates cannot help a chain that has already committed a bad token, whereas extra restarts can. Third — and this addresses the natural objection that a single chain *can* be made revisable — we added the DFM detailed-balance corrector (Campbell et al. 2024): any rate R^{DB} in detailed balance with the conditional marginals can be added at strength η without changing the prior’s marginal flow, which for masking means unmasked coordinates re-mask at rate η while the unmask rate grows to $(\dot{\kappa} + \eta\kappa)/(1 - \kappa)$. In the toy this lifts MC rate guidance to 5/5 recovery ($\eta=8$). At image scale its effect is *symmetric in the wrong way*: editing improves ($0.165 \rightarrow 0.143$, the best non-SMC editing result — revision does help escape a bad source) but recovery *degrades* ($0.111 \rightarrow 0.136$), because the same uniform churn that erases a bad initialization erases a good one — at $\eta=8$ the recovery and editing chains become nearly indistinguishable, the initialization having been fully forgotten. Detailed-balance exploration is revision without *selection*: it re-masks good and bad tokens at the same rate. SMC’s resampling is the selective version of the same operation — states die *because* of their loss — and that is the mechanism a single guided chain has no analogue of. We then tested whether that selectivity can be retrofitted at zero oracle cost: a *regret-guided* remask keeps the same total churn as uniform $\eta=8$ but allocates it by softmax of each coordinate’s commit regret — the mean loss of the decodes that chose its committed token minus the mean over all refills, both cached for free from the K decodes at its commit step — so tokens that underperformed their own refill distribution re-mask preferentially (rate-ratio guidance applied to the remask jumps). It preserves the toy’s 5/5 recovery, but at image scale it is indistinguishable from uniform churn: editing identical (0.143), recovery marginally better ($0.136 \rightarrow 0.131$) yet still worse than no revision at all (0.111), and the recovery and editing trajectories still coincide. The failure localizes the problem precisely: the regret statistics are *stale* — they score a token against the canvas as it stood at commit time, not as it stands now — whereas resampling always scores whole *current* states. Selective single-chain revision would need fresh counterfactual twist estimates per unmasked coordinate at every step, exactly the cost that resampling amortizes across a population. Consistent with this, TFG-Flow’s own benchmarks live in small alphabets (DNA: 4, proteins: 20), where per-step decodes cover the space and the toy result shows the method at its best. The closed-form arm is not useless —

one seed produced “an elderly man or young woman smiling” ($\mathcal{L}$ =0.109, demographics 0.63/0.88 vs target 0.60/0.84) at *four* sampling-time oracle calls — but on average the additive assumption costs more than the exactness buys.

4 Findings

1. **Selection guides, gradients propose, correctors revise — and each pays only in its regime** (§3.6): recovery needs only twisted selection; annealing and the corrector matter for far-source editing; the tilt matters when the target demands low-prior vocabulary. Failure modes addressed: coverage (tilt), expressivity (canvas width), irreversibility (corrector), premature collapse (annealing).
2. **Distilled samplers make mixtures unreachable** (§3.3); gradient-free guidance can swap them out at zero memory cost — a structural advantage over backprop-through- f_ϕ methods.
3. **Estimator choice inverts with scale**: the unbiased sampled twist wins in the toy world; the biased-but-coherent mode twist wins with real LMs, at a fraction of the cost.
4. **Pure first-order discrete guidance fails** where nominate-then-exact-score succeeds — relevant to D-CBG-style methods.
5. **Compositional vs. distributional straddling**: CLIP-MMD partially accepts “both modes in one image” as a substitute for “a population mixing two modes”.
6. **Selection over a population survives vocabulary scale; locally tilted rates do not** (§3.7): rate-ratio guidance matches SMC in small alphabets at lower cost, but at 18k-word vocabulary it is dominated at every K /restart split of the budget — the missing ingredient is resampling, not estimate precision, and it cannot be imitated for free: remasking guided by cached commit-time regret is indistinguishable from uniform churn.

5 Limitations and next steps

Most results use 3 seeds on few targets; the corrector’s effective prior is an approximation of the masked-diffusion prior (§2.6); far-source editing stalls above the floor; prompts are optimal but not always fluent (except under LLaDA). Next: more targets (in particular low-prior-vocabulary multimodal targets, where the ablation predicts the tilt becomes load-bearing), a remasking corrector with ReMDM guarantees, and continuous-time formalism for the guidance theory.